

Geraud Nangue Tasse

My life philosophy *"What I cannot create, I do not understand"* - Richard Feynman

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Location: Johannesburg, ZA

A Fellow of the Machine Intelligence and Neural Discovery (MIND) Institute (wits.ac.za/mind)

A Faculty of the Robotics, Autonomous Intelligence and Learning (RAIL) Lab (raillab.org)

A Faculty of the Cognition, Adaptation and Learning (CAandL) Lab (caandl.org)

Education

Ph.D. Computer Science, [University of the Witwatersrand](#), 2021 – 2023.

M.Sc. Computer Science (with distinction), [University of the Witwatersrand](#), 2019.

B.Sc. Hons Computer Science and Mathematics (with distinction), [Rhodes University](#), 2018.

B.Sc. Computer Science, Pure & Applied Maths, Physics & Electronics, [Rhodes University](#), 2015 – 2017.

Summer Schools

CIFAR Deep Learning + Reinforcement Learning Summer School (*Montreal, CA*), 2023

CIFAR Deep Learning + Reinforcement Learning Summer School (*Remotely*), 2020

IBRO-SIMONS Computational Neuroscience Imbizo (*Cape Town, ZA*), 2020

Skills

Artificial intelligence, Machine learning, Reinforcement learning, Python, C, Java

Employment

Lecturer, [University of the Witwatersrand](#), Johannesburg, ZA, 2025 – Present.

Programme coordinator for MSc by Coursework and Research Report in [Computer Science](#) and [Robotics](#).

Supervision of B.Sc. Honours, M.Sc., and Ph.D. students.

Lecturing the "Reinforcement Learning" BSc Honours and MSc course.

Lecturing the "[Computational Intelligence](#)" MSc course covering optimisation methods in general and specific areas like evolutionary algorithms and particle swarm optimisation.

Sessional Lecturer, [University of the Witwatersrand](#), Johannesburg, ZA, 2022 – 2024.

Panel for various M.Sc. research projects.

Supervision of B.Sc. Honours and M.Sc. students.

Programme coordinator for MSc by Coursework and Research Report in [Computer Science](#) and [Robotics](#).

Lecturing the "[Computational Intelligence](#)" MSc course covering optimisation methods in general and specific areas like evolutionary algorithms and particle swarm optimisation.

Building and lecturing a post-graduate course on the "mathematical and statistical foundations of data science".

PhD Research Intern at [IBM](#), New York, US, 2022.

I worked on compositional generalisation in hierarchical reinforcement learning.

Teaching Assistant at [Neuromatch Deep learning Summer School](#), Remotely, 2021.

I was a reinforcement learning project T.A. for a number of project groups. I was recognised for "most groups mentored" and "most timezones covered".

Research Assistant at [RAIL Lab](#), Remotely, 2020.

I worked on extending my MSc work on composition in RL and writing a journal paper on it.

Tutor, [Rhodes University](#), Grahamstown, ZA, 2015 - 2018.

I was an undergraduate computer science 1 and physics 1 tutor; I also graded computer science practicals.

Fellowships and Scholarships and Awards

Fellow of the Machine Intelligence and Neural Discovery (MIND) Institute, 2024 – *Present*

IBM PhD Research Fellowship , 2022 – 2023

Ph.D. Postgraduate Merit Award , 2021 – 2023

Deep Learning Indaba Research Poster Prize , 2019

Deep Learning Indaba-X (ZA) Hackathon Prize , 2019

Open Box Prize for Best Computer Science Honours Student, 2018

Janinne Franke Prize for the Best Computer Science Honours Project , 2018

Dean's list for Academic Merit BSc , 2017

Rhodes-Investec Top 100 Award for Academic Excellence , 2017

Academic Service and Memberships

Selected Presentations

Conference Talk "Introduction to Reinforcement Learning", at [IndabaX](#) in Uganda, Online, 2024.

Lab Talk "Generalisation in Lifelong Reinforcement Learning through Logical Composition", at [Autonomous Agents Research Group](#) in Edinburgh, Online, 2023.

Tutorial "Mathematical Foundations of Machine Learning", at [Deep Learning Indaba](#), Accra, Ghana, 2023.

Lab Talk "Towards Lifelong Reinforcement Learning through Zero-Shot Logical Composition", at [UCT Shock Lab](#) in Cape Town, Online, 2022.

Lab Talk "Towards Lifelong Reinforcement Learning through Zero-Shot Logical Composition", at [Brown Robotics Lab](#) in Providence, Online, 2022.

Conference Talk "Towards Lifelong Reinforcement Learning through Zero-Shot Logical Composition", at [IndabaX](#) in South Africa, ZA, 2022.

Chair and Program Committee and Organiser

Local Chair of [ICRA@4oAfrica](#) at [University of the Witwatersrand](#), Johannesburg, ZA, 2024.

Organiser of [Robot Learning for Africa Workshop](#) at [Deep Learning Indaba](#), Ghana 2023, Senegal 2024.

Reviewing

Conference on Neural Information Processing Systems (NeurIPS)
 International Conference of Machine Learning (ICML)
 International Conference of Learning Representations (ICLR)
 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
 The International Joint Conference on Artificial Intelligence (IJCAI)
 Black in AI workshop at the Conference on Neural Information Processing Systems (NeurIPS)

Memberships

[The Institute of IT Professionals South Africa IITPSA](#), Johannesburg, ZA, 2024 – Present.

Research

Ph.D Computer Science, University of the Witwatersrand, 2021 – 2023.

Keywords: Reinforcement Learning, Lifelong Learning. Supervised by Dr Steven James, Prof Benjamin Rosman

*Title: **Towards Lifelong Reinforcement Learning through Temporal Logics and Zero-Shot Composition.***

This thesis proposes a framework to develop AI agents with three key abilities: Flexibility, Instructability, and Reliability (FIRE). This is achieved by introducing: the logical composition of arbitrary tasks, world value functions for zero-shot skill composition and lifelong sample efficiency, and methods for agents to understand and execute language instructions.

M.Sc Computer Science (with distinction), University of the Witwatersrand, 2019.

Keywords: Reinforcement Learning, Task composition. Supervised by Dr Steven James, Prof Benjamin Rosman

*Title: **A Task Algebra For Agents In Reinforcement Learning.***

This work addresses the problem of zero-shot logical composition by first formally defining the composition of tasks as operators acting on a set of tasks in an algebraic structure. This provides a structured way of doing task compositions and a theoretically rigorous way of studying them. We propose a framework for defining lattice algebras and Boolean algebras in particular over the space of tasks. This allows us to formulate new tasks in terms of the negation, disjunction, and conjunction of a set of base tasks. We then show that by learning a new type of goal-oriented value functions and restricting the rewards of the tasks, an agent can solve composite tasks with no further learning.

B.Sc Hons Computer Science & Mathematics (with distinction), Rhodes University, 2018.

Keywords: Deep Neural Networks, Hyperparameter Optimization. Supervised by James Connan

*Title: **Hypersearch: A Parallel Training Approach For Optimizing Neural Networks Performance.***

This work proposes a novel embarrassingly parallel algorithm, Hypersearch, that takes advantage of both worlds (hyperparameter optimization and parallel programming). Hypersearch works by training multiple neural networks of same architecture but different hyperparameters in parallel while optimizing both networks and hyperparameters. We manage to provide good evidence that Hypersearch not only performs well in realistic settings, but is also competitive against state-of-the-art Hyperparameter optimization alternatives.

Selected Peer-reviewed Journals

Composition and Zero-Shot Transfer with Lattice Structures in Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

Journal of Artificial Intelligence Research (JAIR), 2025.

Compositional Instruction Following with Language Models and Reinforcement Learning.

Vanya Cohen, **Geraud Nangue Tasse**, Nakul Gopalan, Steven James, Matthew Gombolay, Ray Mooney and Benjamin Rosman.

Transactions on Machine Learning Research (TMLR), 2024. [\[URL\]](#)

Selected Peer-reviewed Conferences

Skill Machines: Temporal Logic Skill Composition in Reinforcement Learning.

Geraud Nangue Tasse, Devon Jarvis, Steven James and Benjamin Rosman.

In Proceedings of the Twelfth International Conference on Learning Representations (ICLR), 2024. [\[URL\]](#)

Generalisation in Lifelong Reinforcement Learning through Logical Composition.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In Proceedings of the Tenth International Conference on Learning Representations (ICLR), 2022. [\[URL\]](#)

A Boolean Task Algebra for Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In Advances in Neural Information Processing Systems (NeurIPS), 2020. [\[URL\]](#)

Selected Peer-reviewed Workshops and Symposia

ROSARL: reward-only safe reinforcement learning.

Geraud Nangue Tasse, Tamlin Love, Mark Nemecek, Steven James and Benjamin Rosman.

In Reinforcement Learning Safety Workshop at RLC, 2024. [\[URL\]](#)

Optimal task generalisation in cooperative multi-agent reinforcement learning.

Geraud Nangue Tasse, Tamlin Love, Mark Nemecek, Steven James and Benjamin Rosman.

In Coordination and Cooperation in Multi-Agent Reinforcement Learning Workshop at RLC, 2024. [\[URL\]](#)

End-to-end learning to follow language instructions with compositional policies.

Vanya Cohen, **Geraud Nangue Tasse**, Nakul Gopalan, Steven James, Matthew Gombolay, Ray Mooney and Benjamin Rosman.

In Workshop on Language and Robot Learning at CORL, 2022. [\[URL\]](#)

World Value Functions: Knowledge Representation for Multitask Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM), 2022. [\[URL\]](#)

Learning to follow language instructions with compositional policies.

Vanya Cohen, **Geraud Nangue Tasse**, Nakul Gopalan, Steven James, Matthew Gombolay, Ray Mooney and Benjamin Rosman.

In AAAI Fall Symposium on AI for Human-Robot Interaction, 2021. [\[URL\]](#)

Logical composition for Reinforcement Learning.

Geraud Nangue Tasse, Steven James and Benjamin Rosman.

In 4th Life-long Learning Workshop at the International Conf. on Machine Learning (ICML), 2020. [\[URL\]](#)

Hypersearch: A Parallel Training Approach For Improving Neural Networks Performance.

Geraud Nangue Tasse and James Connan.

Black in AI workshop at the 33rd Conference on Neural Information Processing Systems (NeurIPS), 2019.